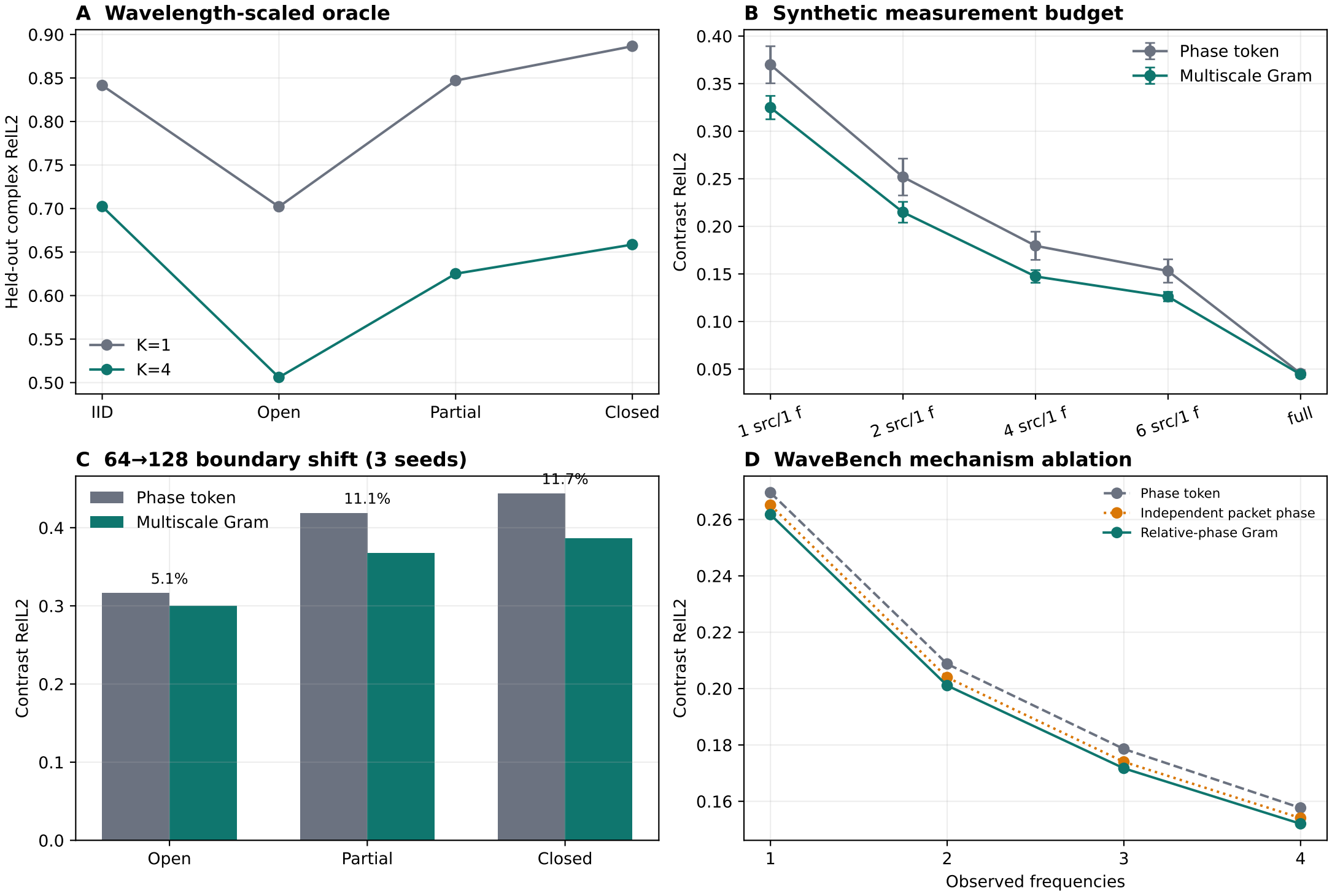
A Wavelength-scaled oracle
Held-out complex RelL2
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
K=1
K=4
IID
Open
Partial
Closed
B Synthetic measurement budget
Contrast RelL2
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
Phase token
Multiscale Gram
1 src/1 f
2 src/1 f
4 src/1 f
6 src/1 f
full
C 64→128 boundary shift (3 seeds)
Contrast RelL2
0.4
0.3
0.2
0.1
0.0
Phase token
Multiscale Gram
5.1%
11.1%
11.7%
Open
Partial
Closed
D WaveBench mechanism ablation
Contrast RelL2
0.26
0.24
0.22
0.20
0.18
0.16
Phase token
Independent packet phase
Relative-phase Gram
1
2
3
4
Observed frequencies